Clutches and Brakes

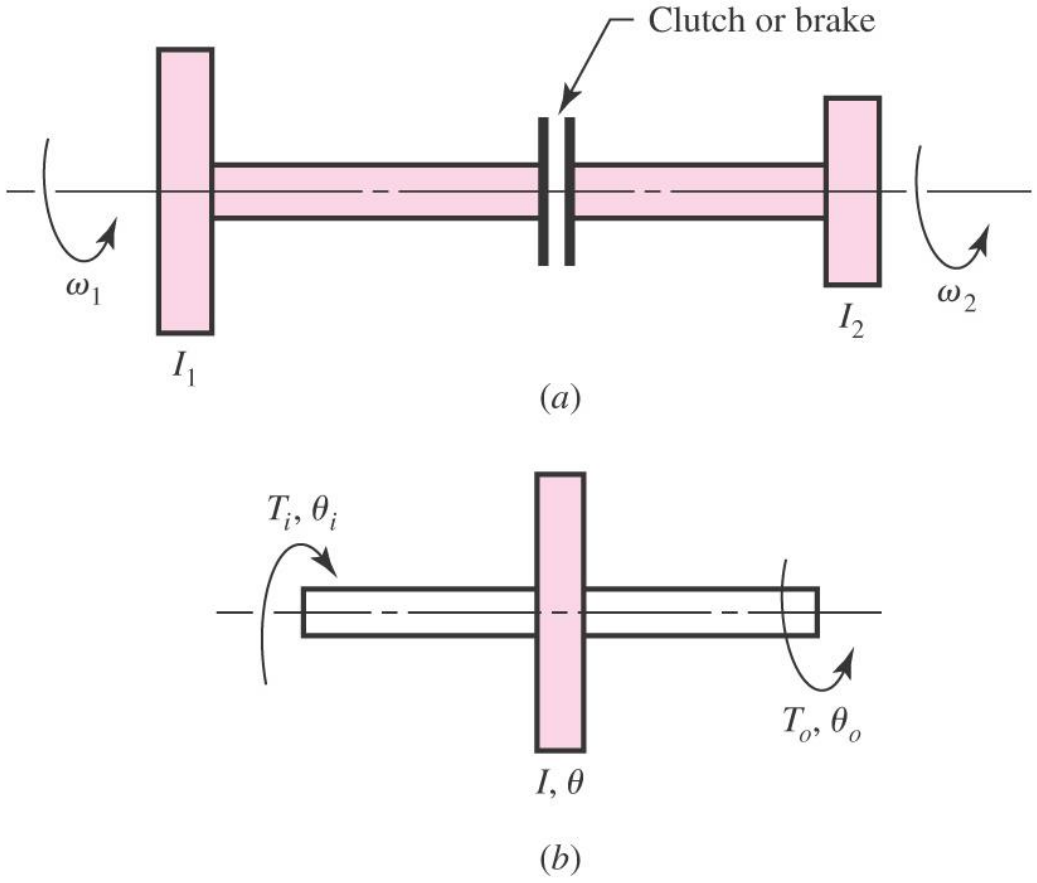
Prof. S. P. Owusu-Ofori

Based on Machine Component Design by Juvinall

Clutch versus Brake versus Flywheel

Function of Clutch/Brake:

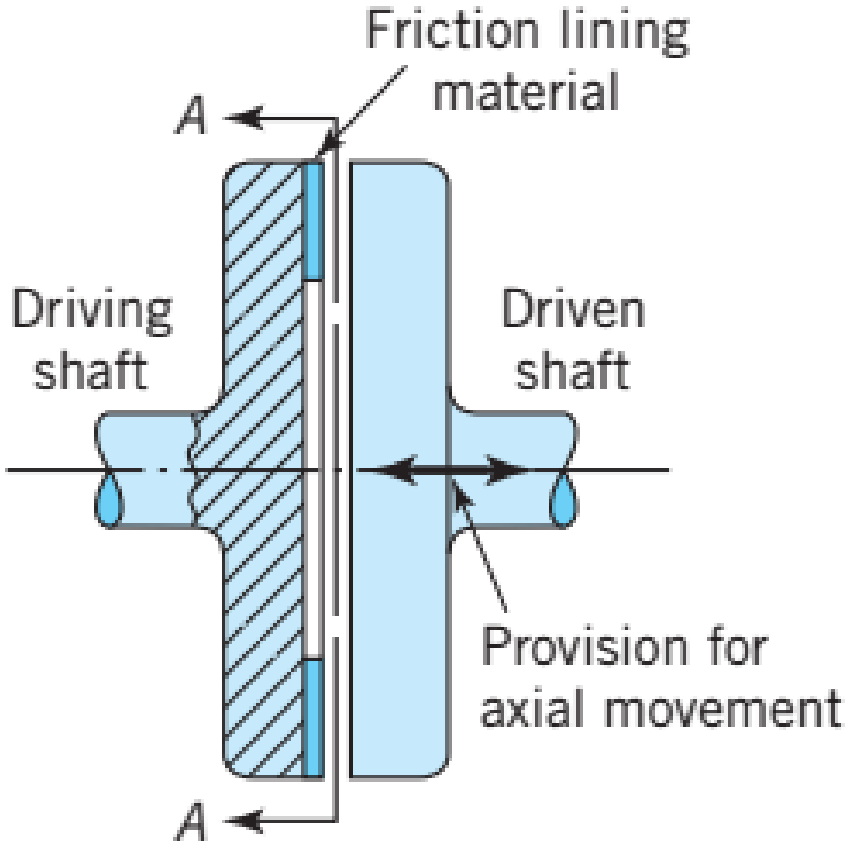
- Permits smooth, gradual connection and disconnection of two members having a common axis.
- Transfers rotational energy from one component to the other.

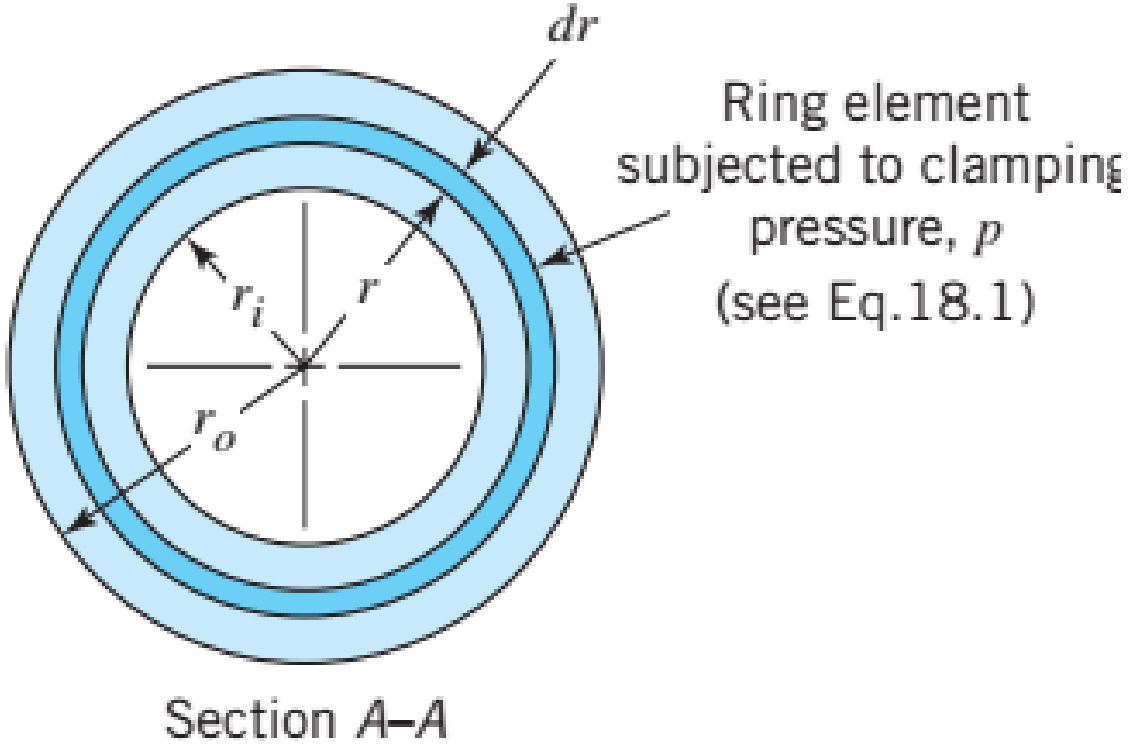


Function of Flywheel:

- Inertia storage device.
- Absorbs mechanical energy increasing its speed and dissip the energy by decreasing its speed.

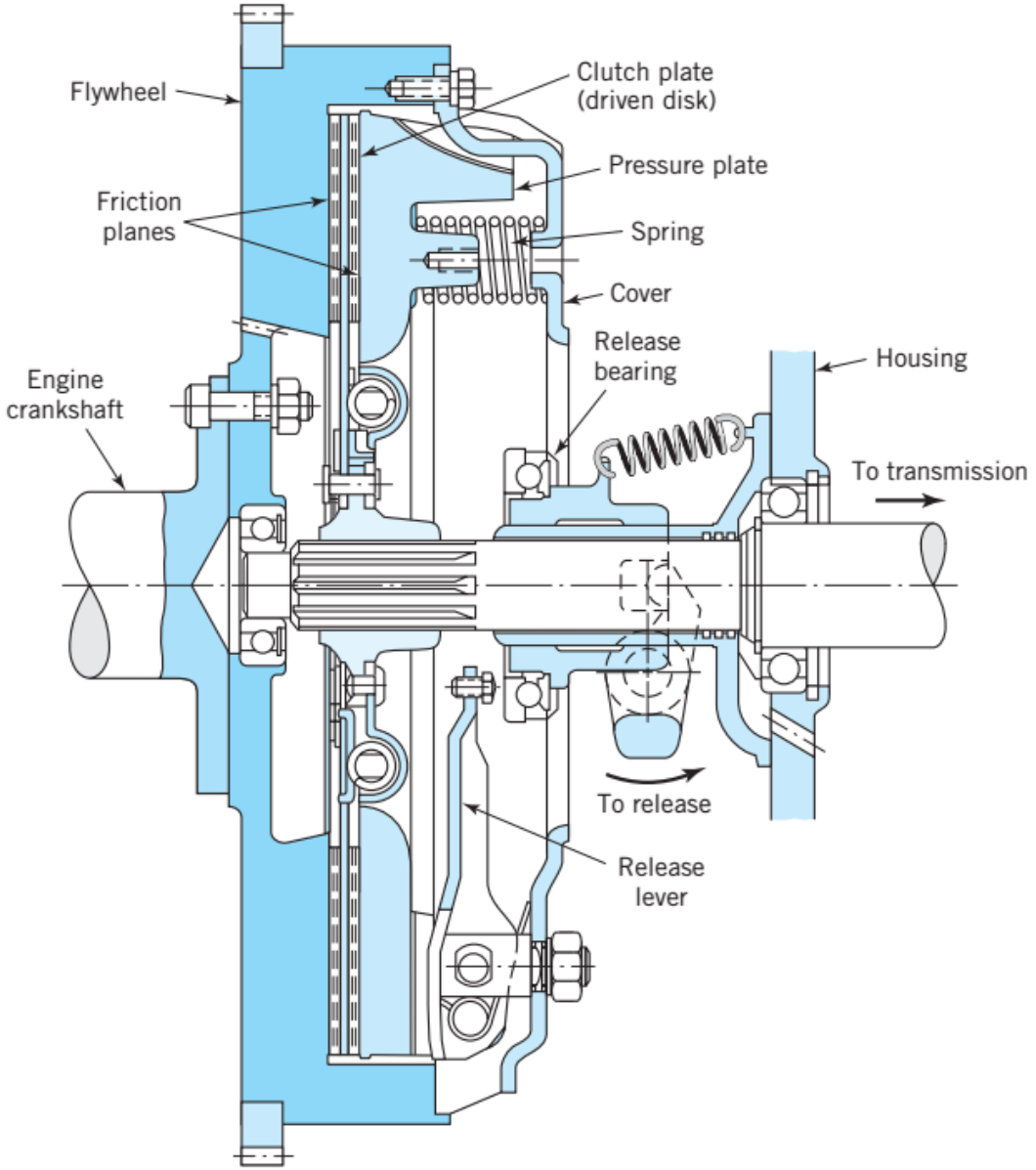
Typical Disk Clutch



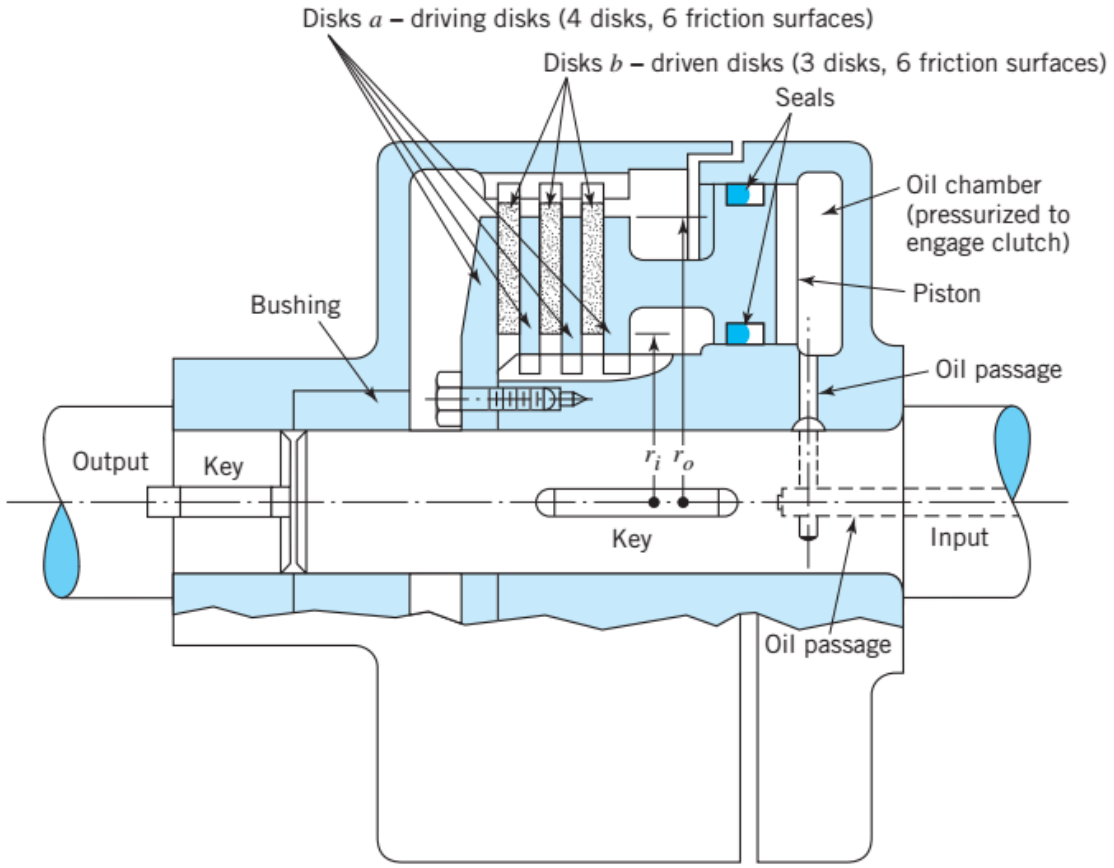


Automotive Disk Clutch

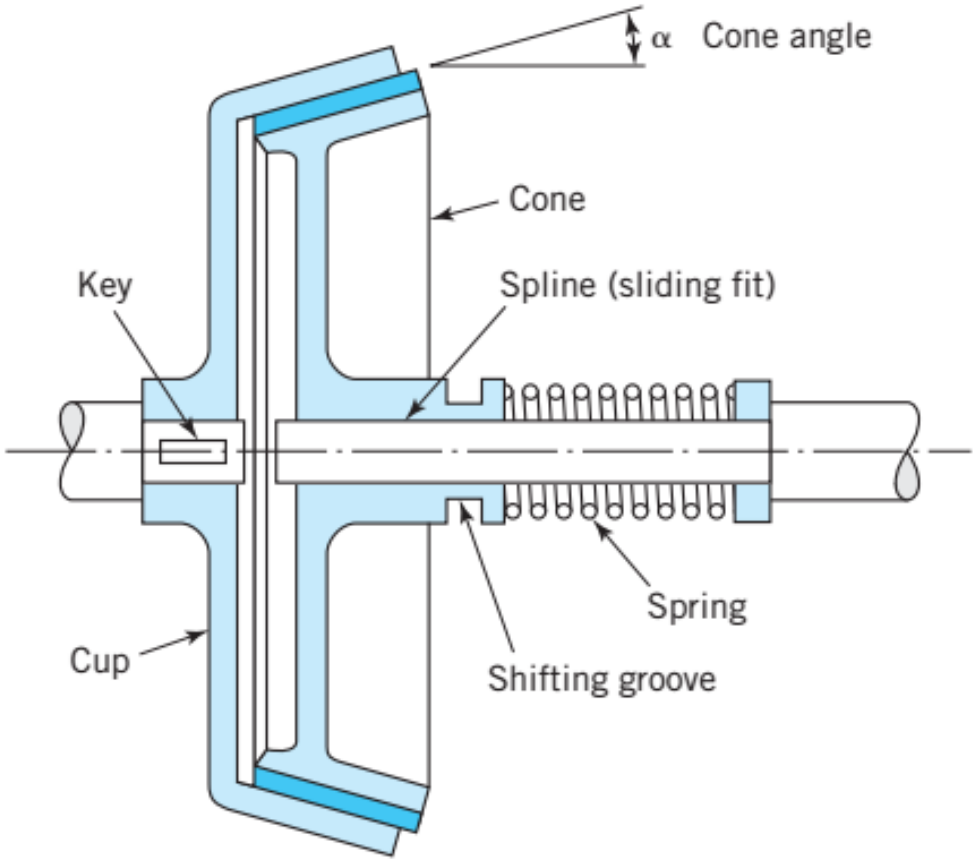
Automotive
Disk Clutch



Multi-disk and Cone Clutches



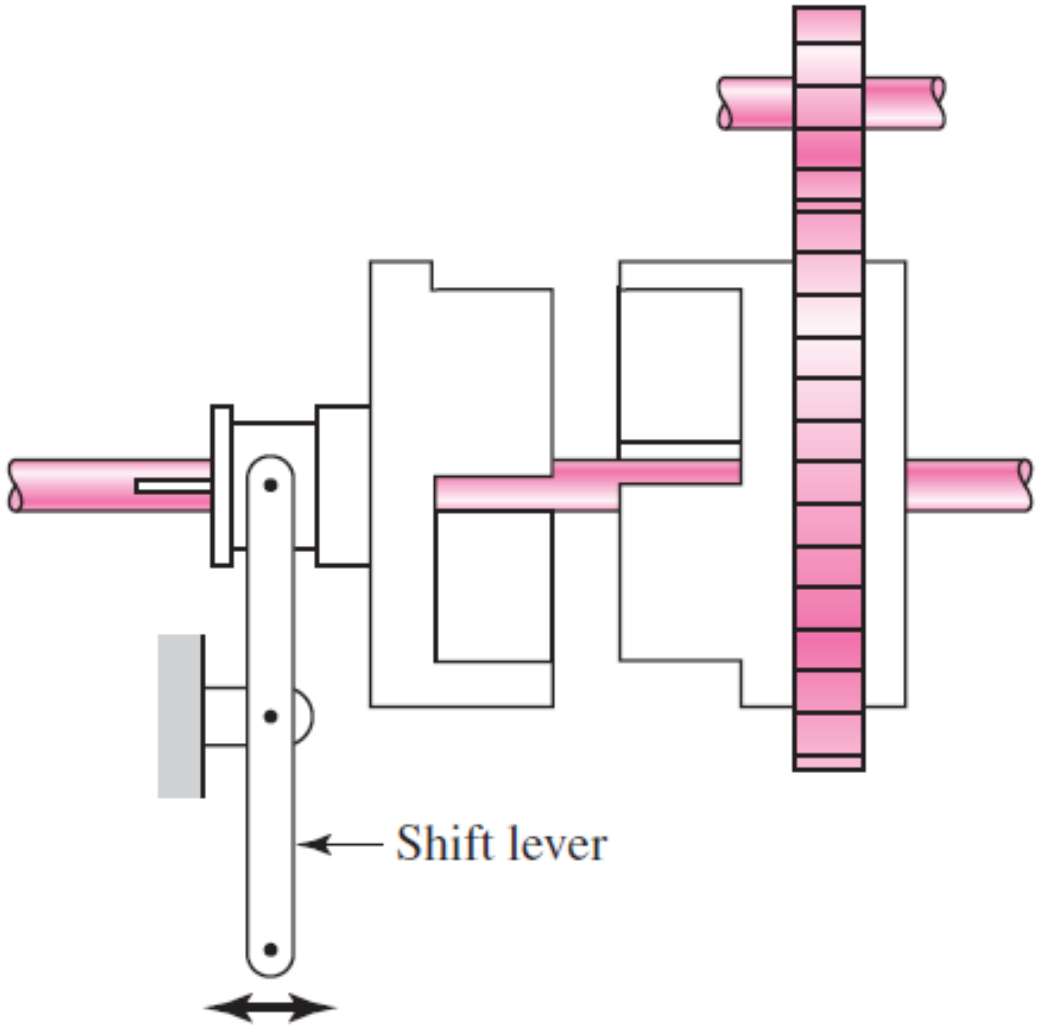
Multiple-Disk Clutch



Cone Clutch

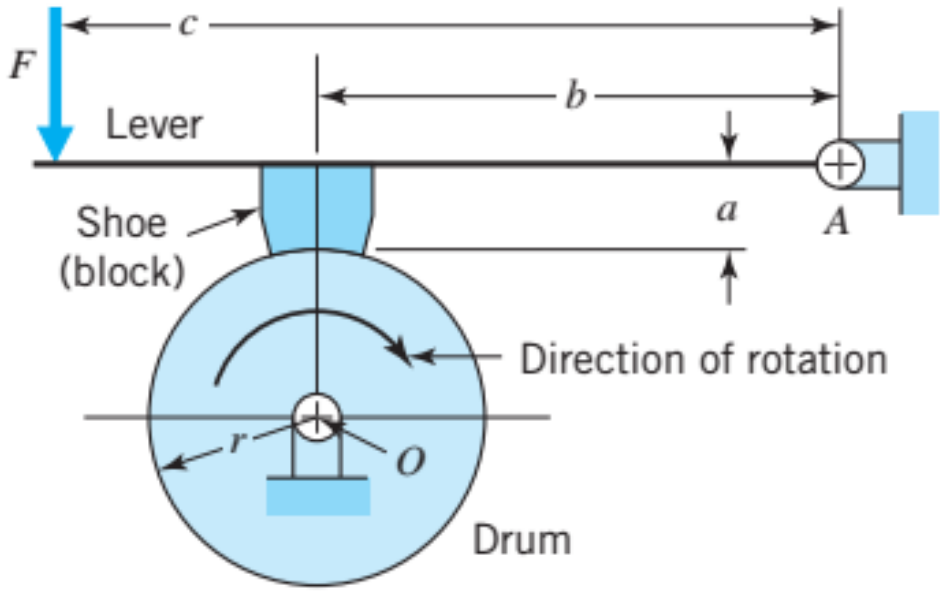
Positive-Contact/Drive Clutches

- Characteristics of positive-contact clutches
  - No slip
  - No heat generated
  - Cannot be engaged at high speeds
  - Sometimes cannot be engaged when both shafts are at rest
  - Engagement is accompanied by shock

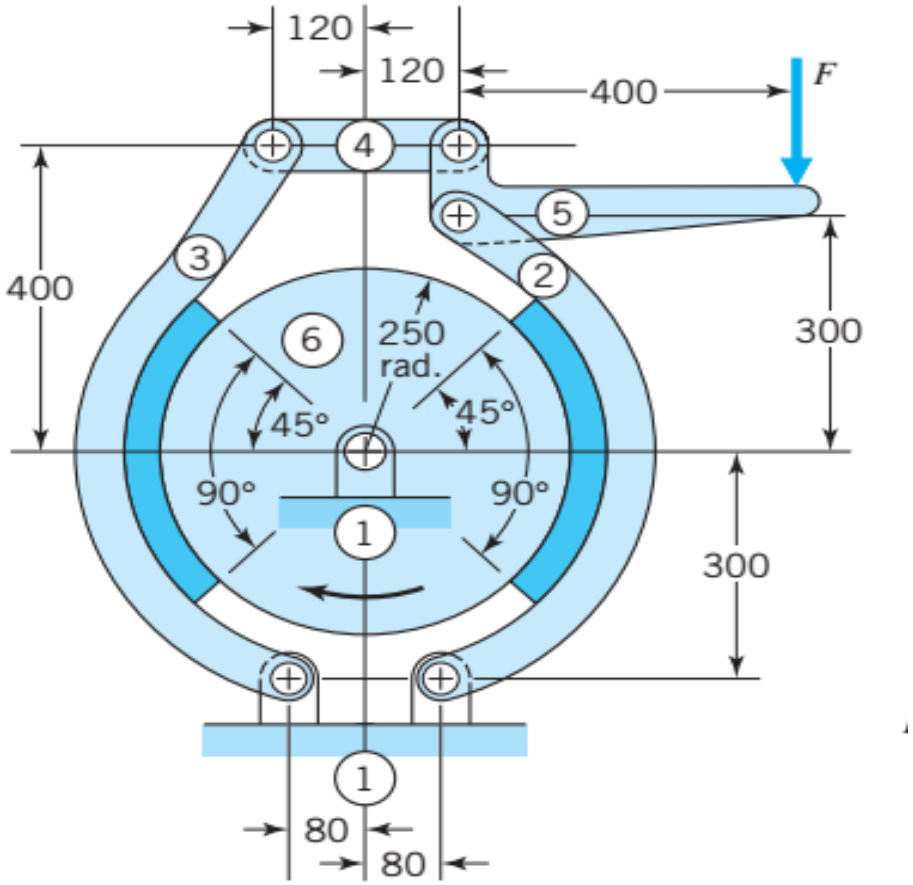


Square-jaw Clutch

Types of Brakes

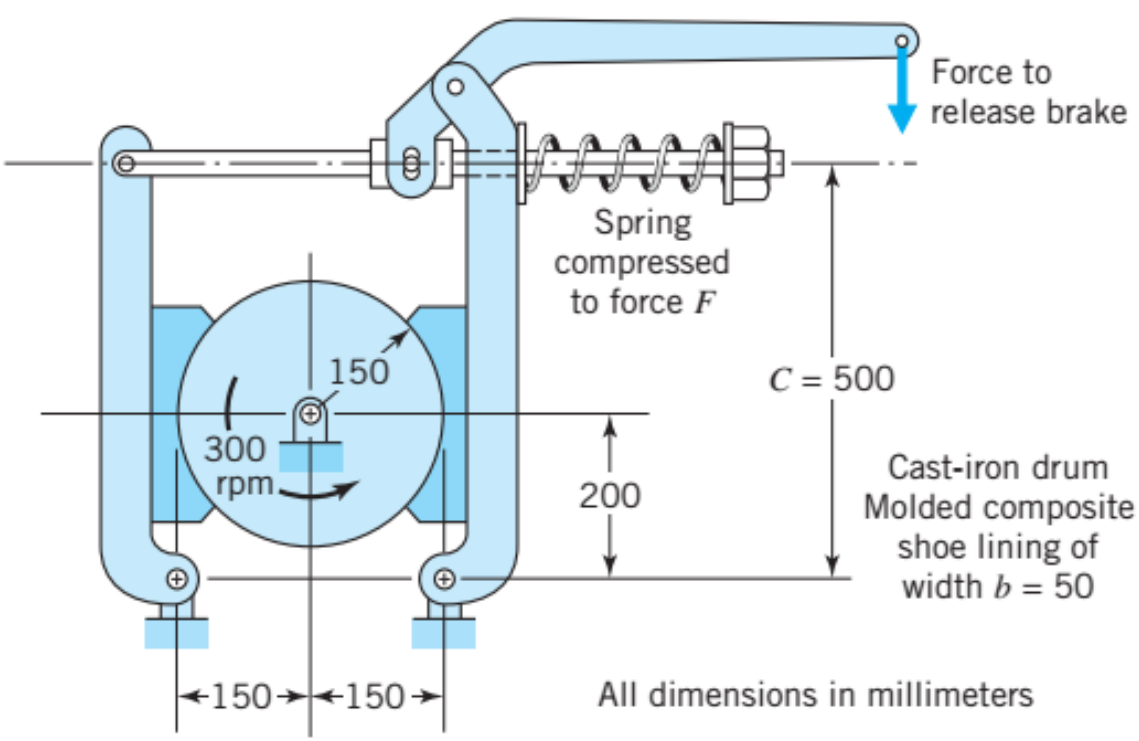


Short-Shoe Drum Brake

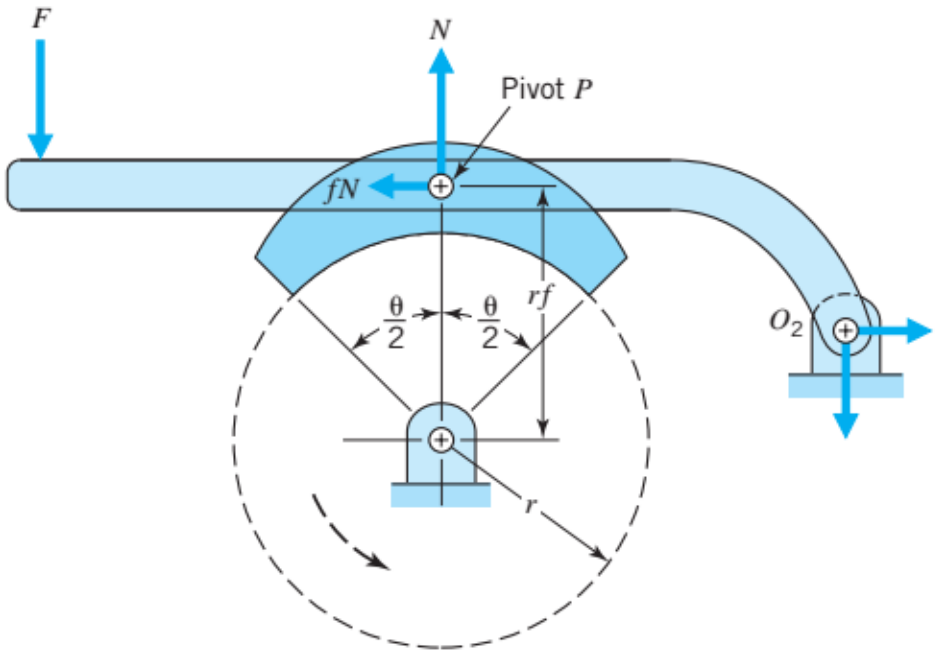


Two-Shoe External
Drum Brake

Types of Brakes



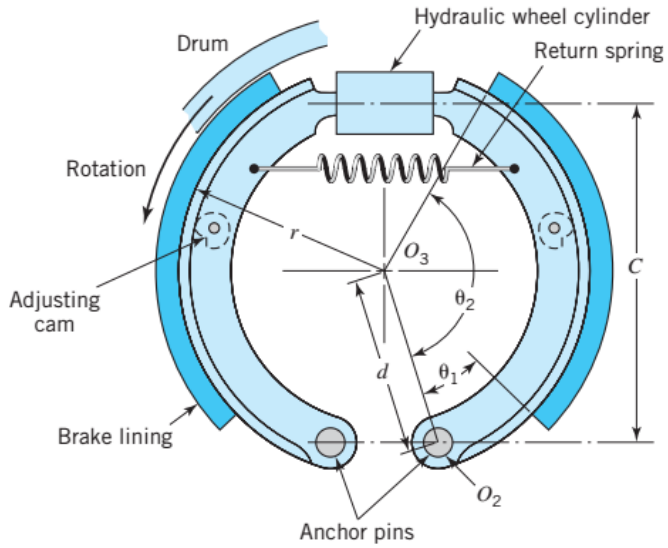
Double-Shoe External Drum Brake



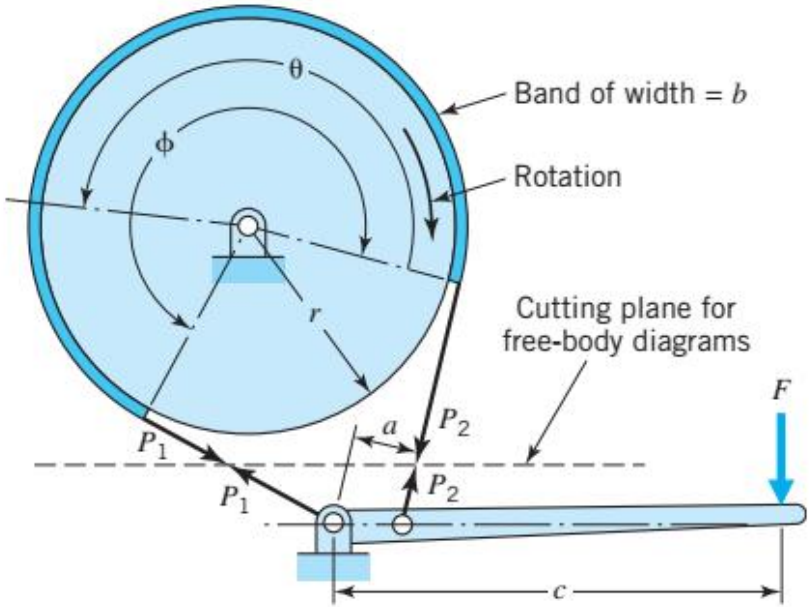
Pivoted Long Shoe

Types of Brakes

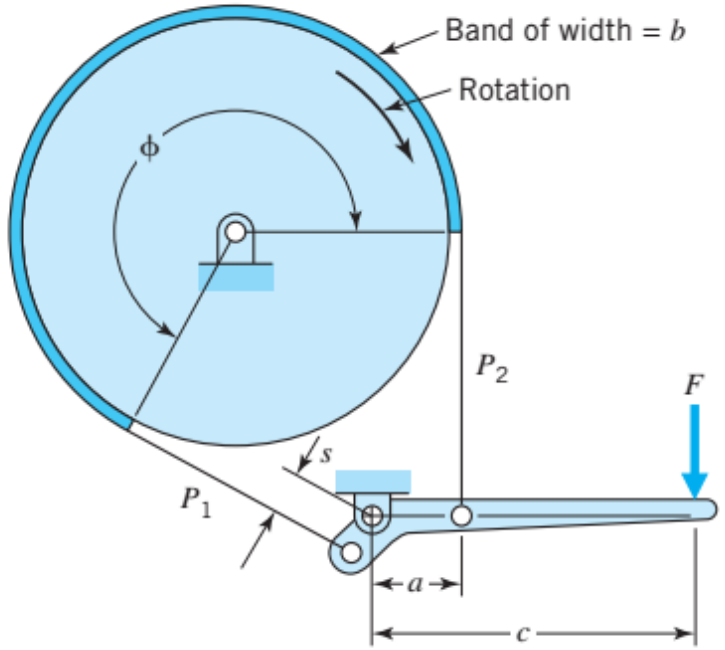
Internal Long-Shoe Drum Brake



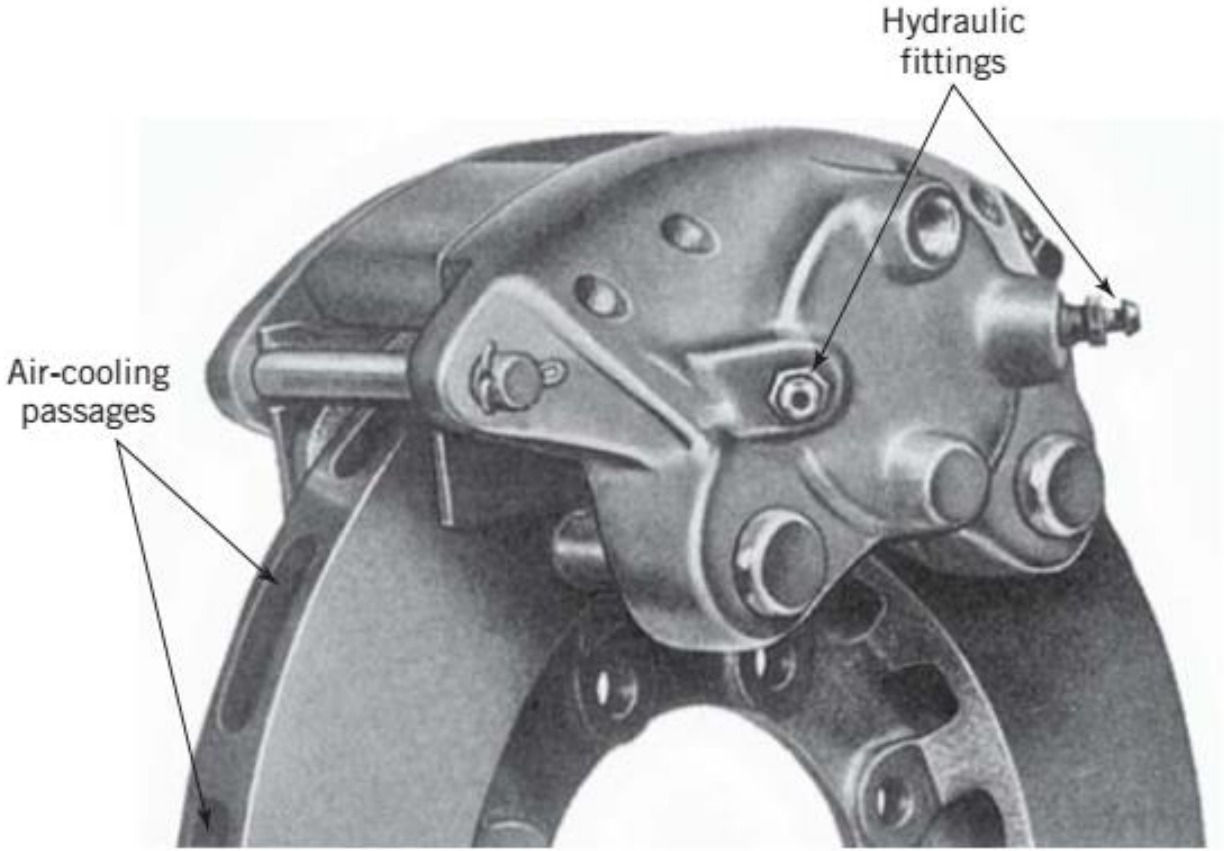
Band Brake



Differential Band Brake



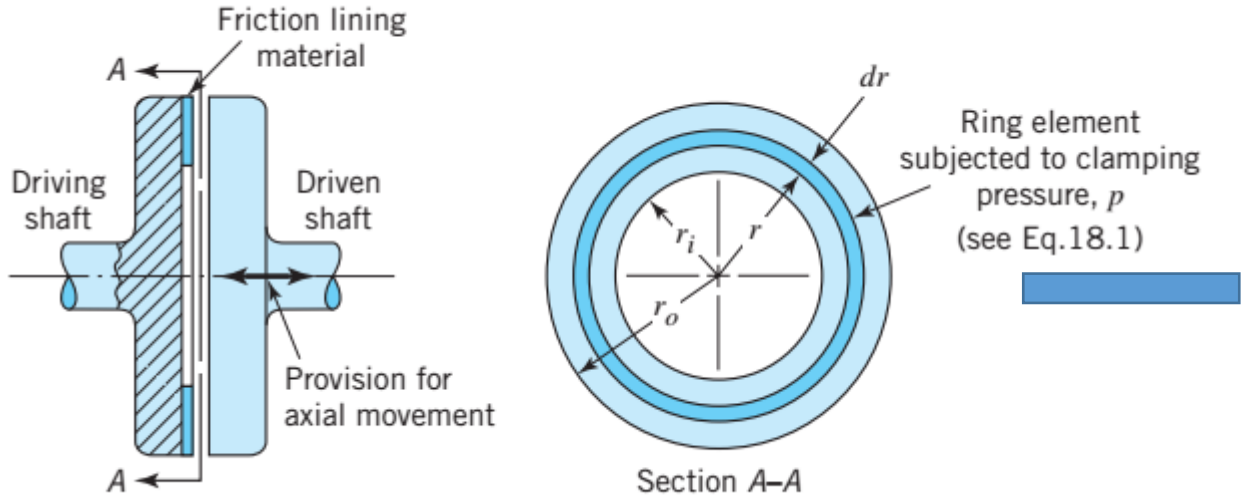
Caliper Disk Brake



Design Performance/Concerns

- **Actuating Force** – Force need to bring the two elements together
- **Torque Transmitted** – Related to the actuating force, coefficient of friction, and geometry
- **Temperature Rise/Friction Energy** – due to friction caused by slippage between the two elements when the speeds are different
- **Wear Rate** – characteristics of friction surface must give acceptable life

Analysis of Disk Clutches



a. UNIFROM PRESSURE DISTRIBUTION

$$dF = (2\pi r\, dr)p \longrightarrow F = \int_{r_i}^{r_o} 2\pi pr\, dr = \pi p(r_o^2 - r_i^2)$$

$$dT = (2\pi r\, dr)pfr \longrightarrow T = \int_{r_i}^{r_o} 2\pi pfr^2\, dr = \tfrac{2}{3}\pi pf(r_o^3 - r_i^3) \longrightarrow T = \tfrac{2}{3}\pi pf(r_o^3 - r_i^3)N$$

$$T = \frac{2Ff(r_o^3 - r_i^3)}{3(r_o^2 - r_i^2)}N$$

In Terms of force F

b. UNIFORM WEAR RATE AT INTERFACE

- Wear Rate is proportional to rate of friction work or friction energy
- Friction energy is proportional to pressure (p) and interface velocity (v)
- Velocity (v) is proportional to the radius, r
- Thus Wear Rate is thus proportional to "pr"
- For uniform wear rate, ***pr = constant (C)***
- ***Show that p_{max} occurs at r_i NOTE: p_{max} is normally specified for the clutch***
- ***Then $pr = p_{max} r_i$***

$$F = \int_{r_i}^{r_o} 2\pi p_{\max} r_i \, dr = 2\pi p_{\max} r_i (r_o - r_i)$$

$$T = \int_{r_i}^{r_o} 2\pi p_{\max} r_i f r \, dr \, N = \pi p_{\max} r_i f (r_o^2 - r_i^2) N$$

OR

$$T = Ff\left(\frac{r_o + r_i}{2}\right)N$$

Maximum torque occurs when:

$$r_i = \sqrt{\tfrac{1}{3}} r_o = 0.58 r_o$$

Proportions commonly used range from $r_i = 0.45 r_o$ to $r_i = 0.80 r_o$.

Analysis of Disk Clutches

TABLE 18.1 Representative Properties of Friction Materials, Operating Dry

Friction Material[a]	Dynamic Friction Coefficient f[b]	Maximum Pressure[c]		Maximum Bulk Temperature	
		psi	kPa	°F	°C
Molded	0.25–0.45	150–300	1030–2070	400–500	204–260
Woven	0.25–0.45	50–100	345–690	400–500	204–260
Sintered metal	0.15–0.45	150–300	1030–2070	450–1250	232–677
Cork	0.30–0.50	8–14	55–95	180	82
Wood	0.20–0.30	50–90	345–620	200	93
Cast iron, hard steel	0.15–0.25	100–250	690–1720	500	260

[a] When rubbing against smooth cast iron or steel.
[b] Experimental values of f vary with detailed composition, rubbing speed, pressure, temperature, humidity, and composition. Consult the manufacturer or obtain test data. For design, commonly use 50 to 75 percent of test values to provide a factor of safety.
[c] Use of lower values will give longer life. Consult the manufacturer or obtain test data. Compute the average pressure on cylindrical surfaces on the basis of projected area of contact (as for bearing stresses and journal bearing pressures—see Section 13.3).

Friction Coefficients

TABLE 18.2 Representative Values of Friction Coefficient for Friction Materials Operating in Oil

Friction Material[a]	Dynamic Friction Coefficient f
Molded	0.06–0.09
Woven	0.08–0.10
Sintered metal	0.05–0.08
Paper	0.10–0.14
Graphitic	0.12 (avg.)
Polymeric	0.11 (avg.)
Cork	0.15–0.25
Wood	0.12–0.16
Cast iron, hard steel	0.03–0.06

[a] When rubbing against smooth cast iron or steel.

Example 1 – Disk Clutch

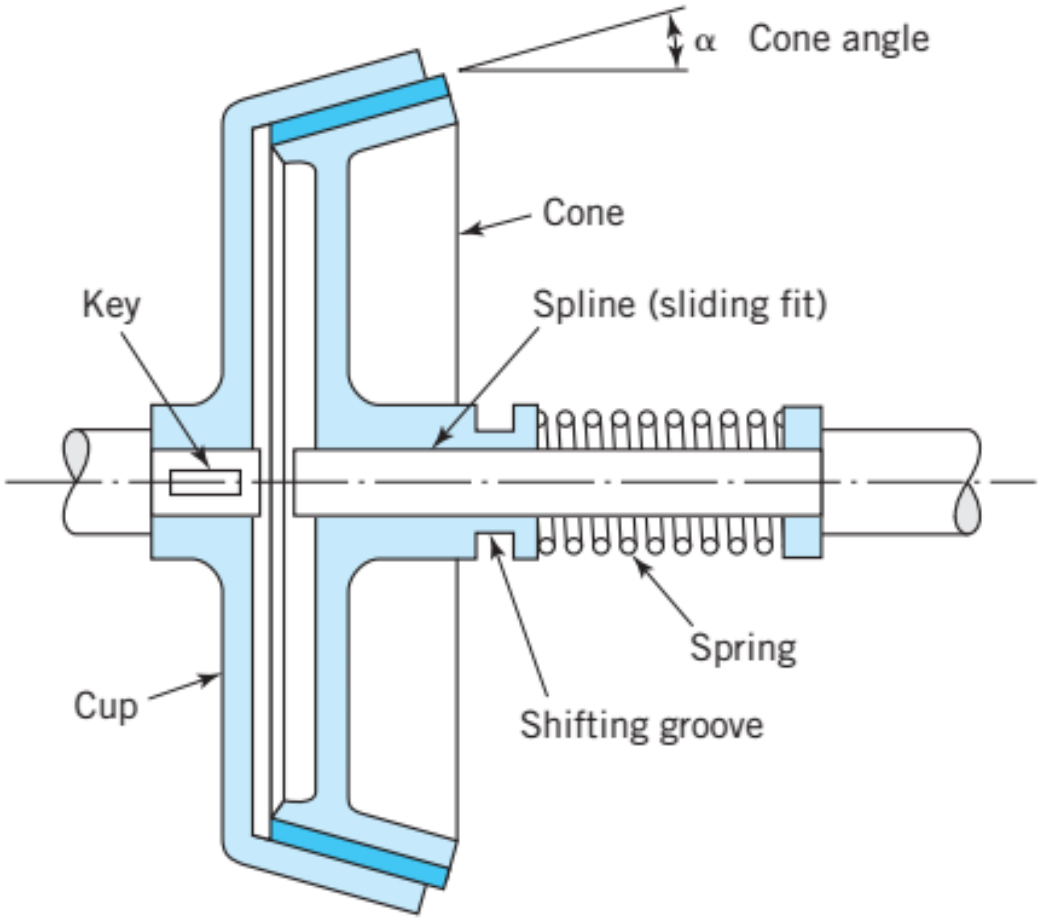
A multiple-disk wet clutch is to be designed for transmitting a torque of 85 N·m. Space restrictions limit the outside disk diameter to 100 mm. Design values for the molded friction material and steel disks to be used are $f = 0.06$ (wet) and $p_{max} = 1400$ kPa. Determine appropriate values for the disk inside diameter, the total number of disks, and the clamping force.

Problem 1 – Disk Clutch

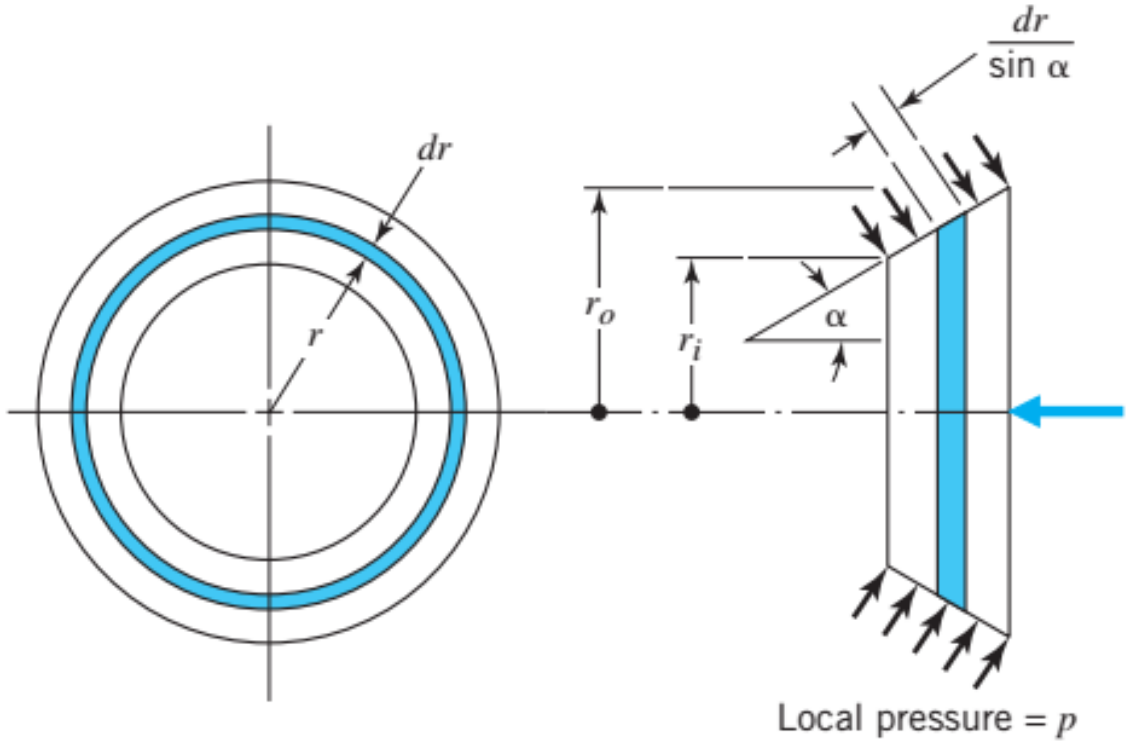
A multiple-disk clutch is to operate in oil and be able to transmit a design overload torque of 800 N · m. The disks are alternately high-carbon steel and molded asbestos, with inside and outside diameters of 90 and 150 mm, respectively. Design values based on test experience for this application are $p_{max} = 1000$ kPa and $f = 0.10$. What total number of disks is required?

[Ans.: 17 (8 disks in one set and 9 in the other, giving 16 friction interfaces)]

Cone Clutches and Brakes



(a)



(b)

Analysis of Cone Clutch and Brake

A. UNIFORM PRESSURE

$$T = \frac{2}{3}\pi pf(r_o^3 - r_i^3)/\sin\alpha$$

$$T = \frac{2Ff(r_o^3 - r_i^3)}{3(r_o^2 - r_i^2)} \Big/ \sin\alpha$$

B. UNIFORM WEAR RATE

$$T = \pi p_{\max} r_i f(r_o^2 - r_i^2)/\sin\alpha$$

$$T = Ff\left(\frac{r_o + r_i}{2}\right) \Big/ \sin\alpha$$

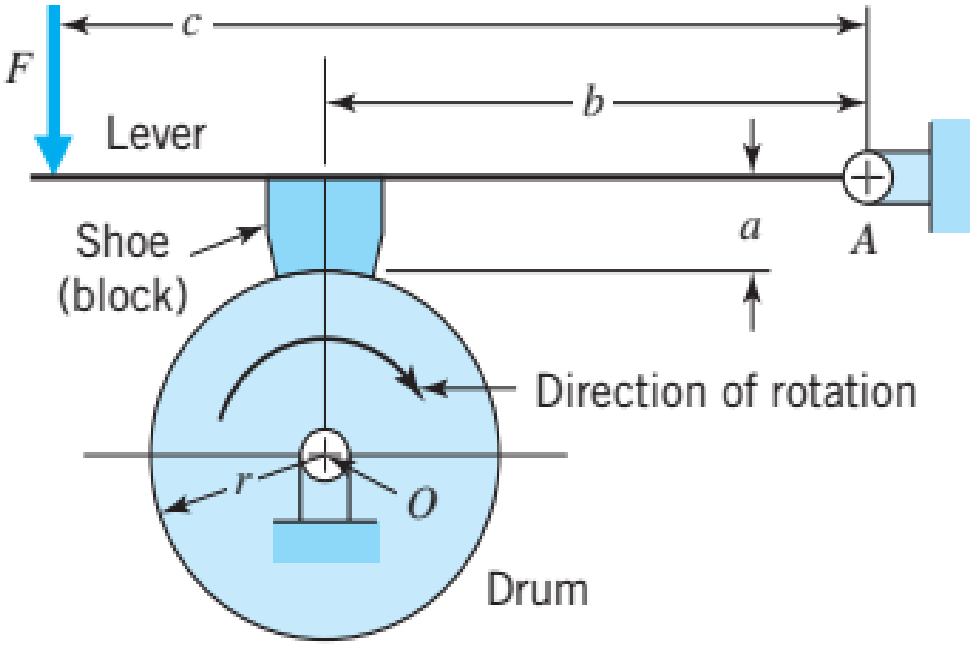
Problem – Cone Clutch

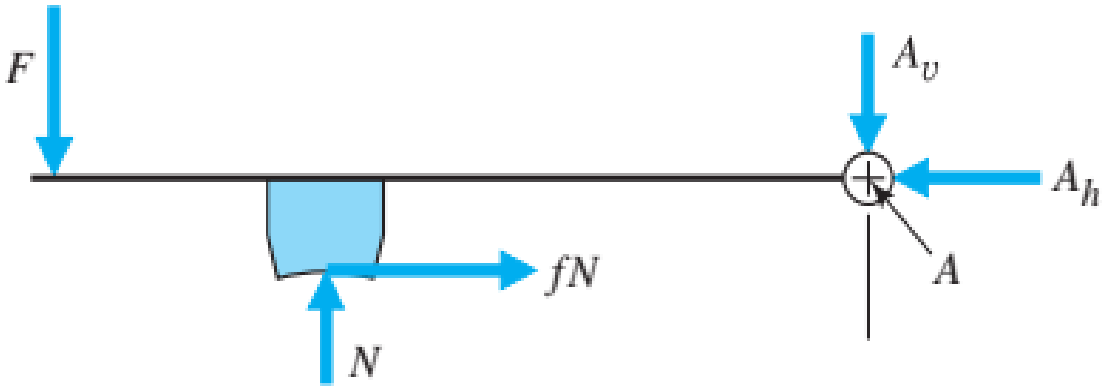
An engine developing 45 kW at 1000 rpm is fitted with a cone Clutch built inside the flywheel. The cone has a face angle of 12.5^{o} and a mean diameter of 500 mm. The coefficient of friction is 0.2. The normal pressure on the clutch face is not to exceed 0.1 N/mm^{2}

Determine: (a) face width required

(b) the axial force necessary to engage the clutch.

Analysis of Short-Shoe Drum Brakes





(b)
Shoe and lever as a free body

Example – Short Show Brake

The following data apply to a short shoe drum brake:

Radius of drum = 125 mm
Shoe width = 50 mm Shoe Length = 100 mm
Parameters: a = 40 mm b= 150 mm c = 250 mm
Pressure, P = 0.7 MPa f = 0.3

a. Determine if the system will self-energize
b. The actuating force, F.

Analysis of Band Brakes

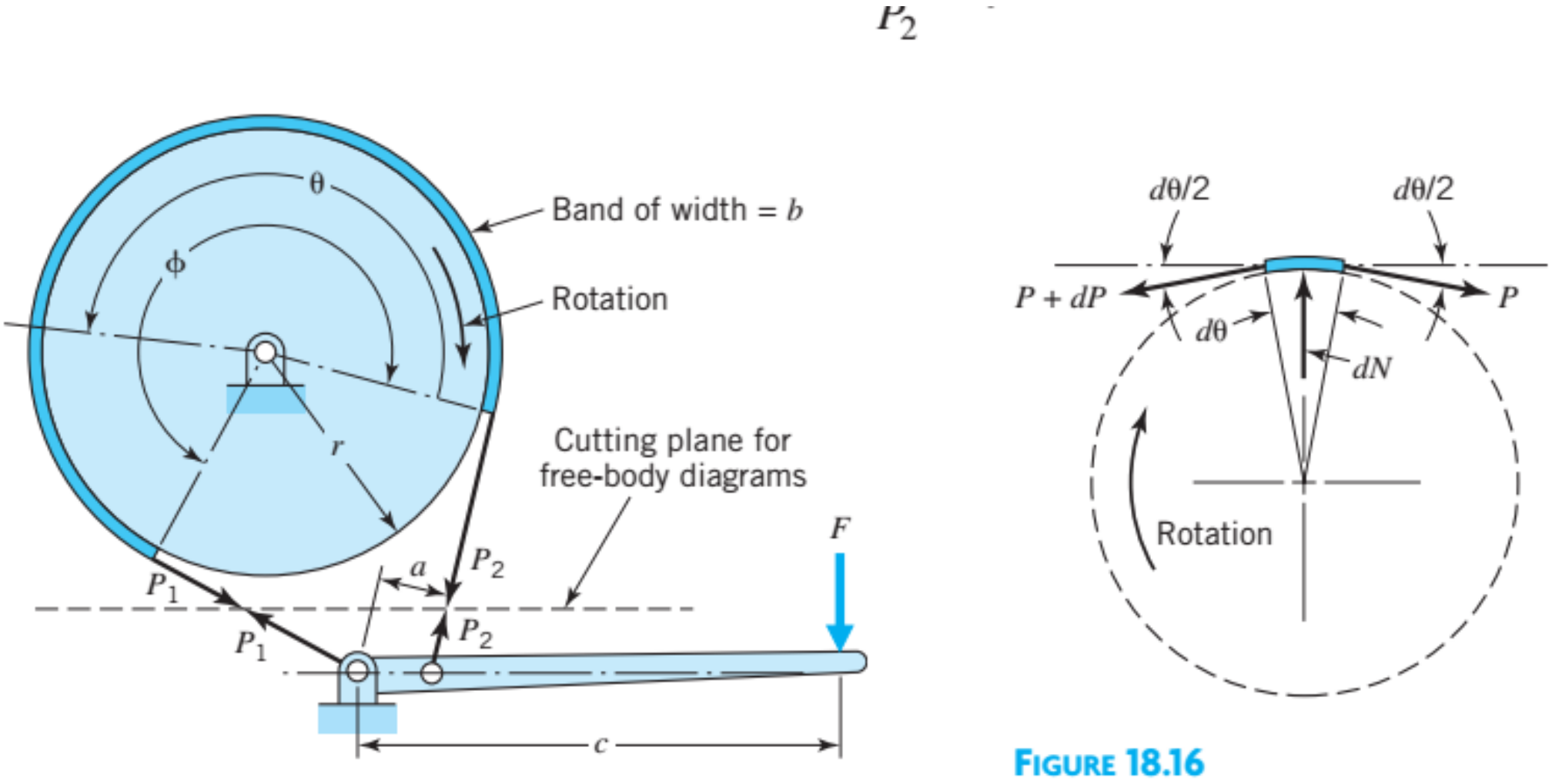
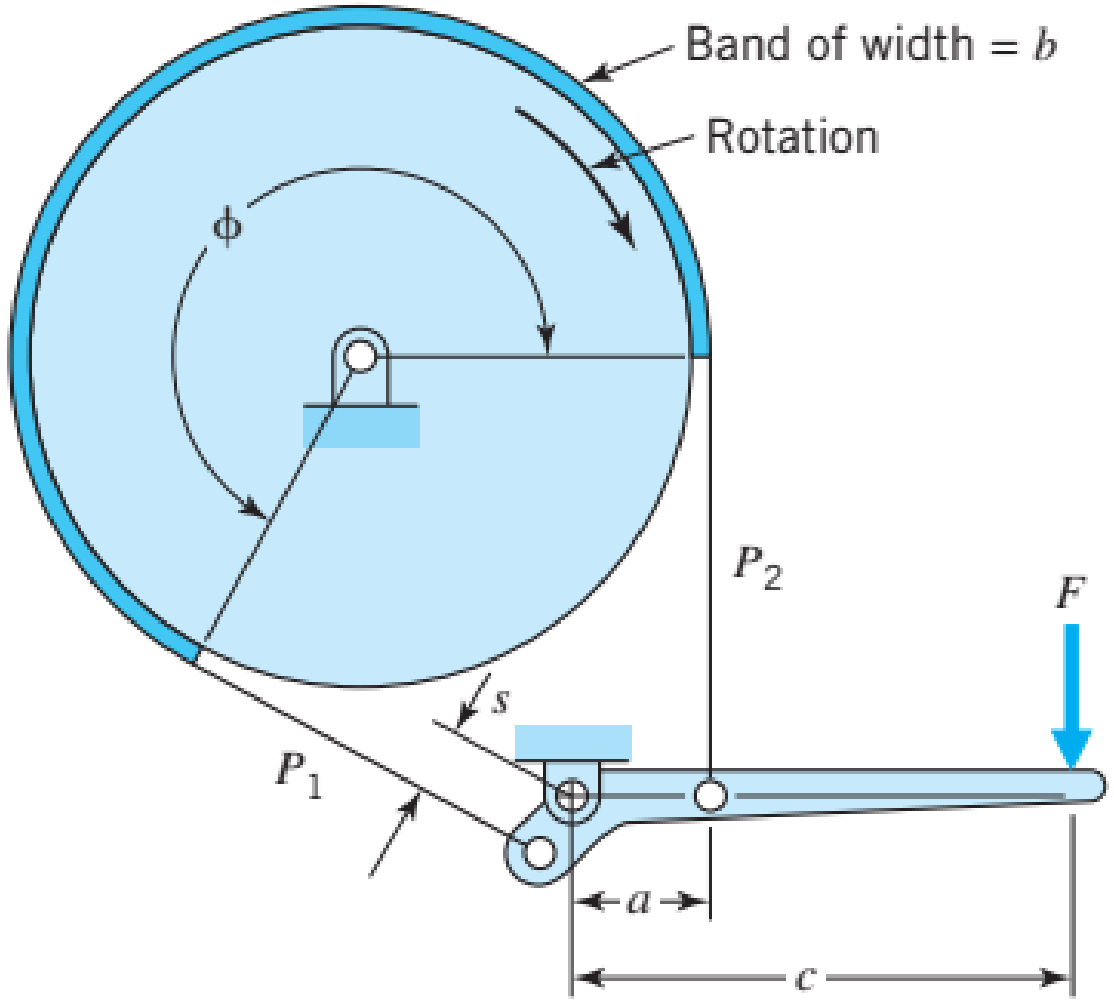


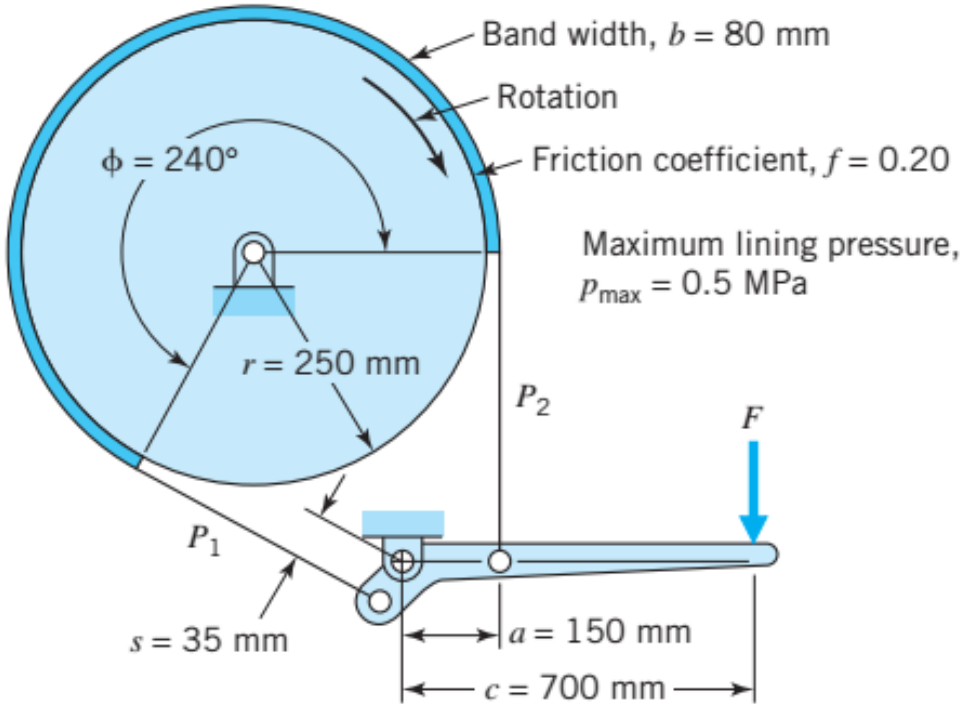
FIGURE 18.16

Analysis of Differential Band Brake



Example – Differential Band Brake

A differential band brake shown in Figure 18.18 uses a woven lining having a design value of $f = 0.20$. Dimensions are $b = 80$ mm, $r = 250$ mm, $c = 700$ mm, $a = 150$ mm, $s = 35$ mm, and $\phi = 240°$. Find (a) the brake torque if the maximum lining pressure is 0.5 MPa, (b) the corresponding actuating force F, and (c) the values of dimension s that would cause the brake to be self-locking.



Temperature Rise

$$\Delta T = \frac{H}{C_p W} \tag{16–54}$$

where ΔT = temperature rise, °F

C_p = specific heat capacity, Btu/(lbm · °F); use 0.12 for steel or cast iron

W = mass of clutch or brake parts, lbm

$$\Delta T = \frac{E}{C_p m} \tag{16–55}$$

where ΔT = temperature rise, °C

C_p = specific heat capacity; use 500 J/kg · °C for steel or cast iron

m = mass of clutch or brake parts, kg

Newton's Cooling Model

$$\frac{T - T_\infty}{T_1 - T_\infty} = \exp\left(-\frac{\hbar_{CR} A}{W C_p} t\right) \qquad (16\text{–}56)$$

where T = temperature at time t, °F

T_1 = initial temperature, °F

T_∞ = environmental temperature, °F

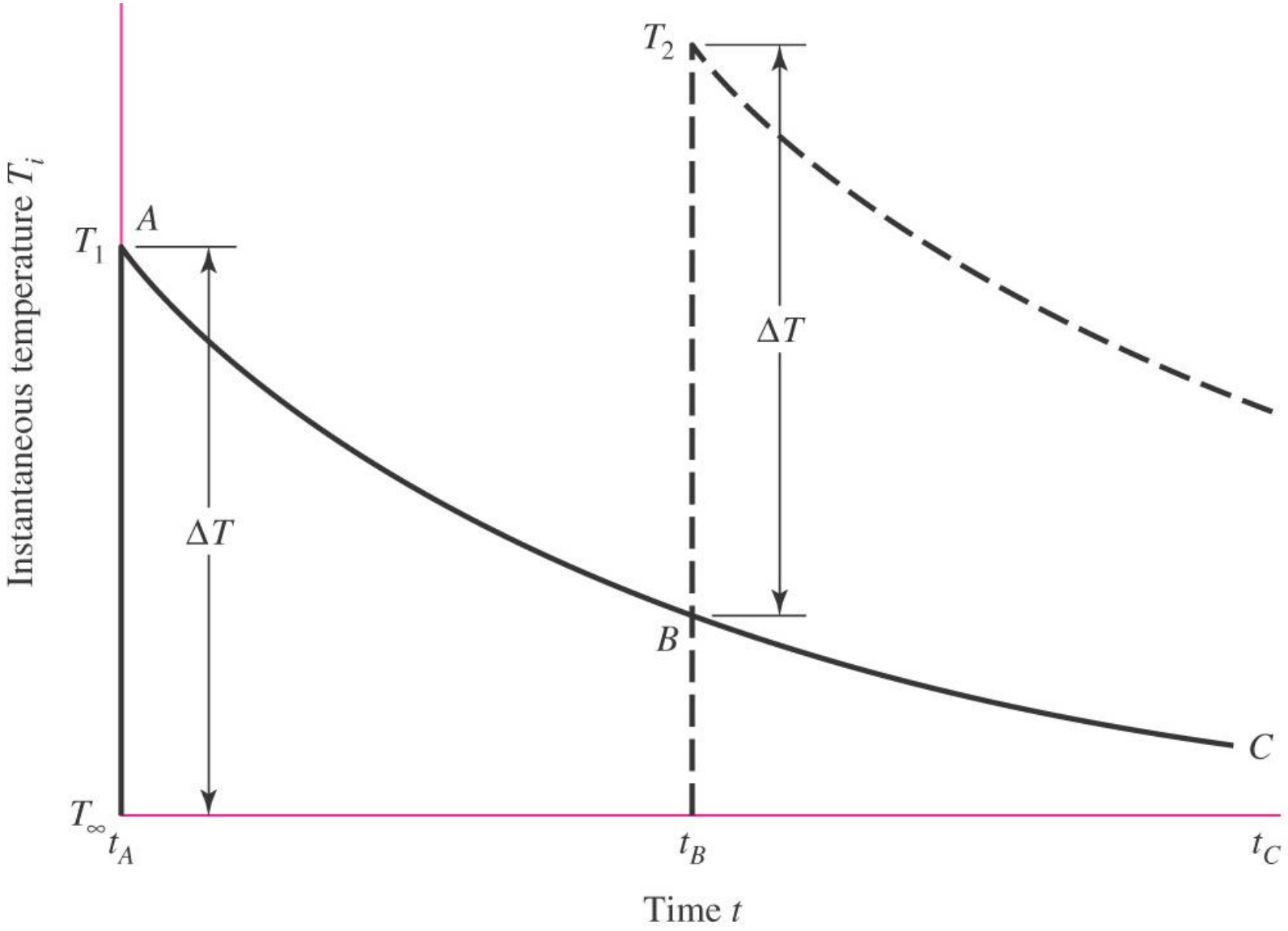
$\hbar_{CR}$ = overall coefficient of heat transfer, Btu/(in^2 · s · °F)

A = lateral surface area, in^2

W = mass of the object, lbm

C_p = specific heat capacity of the object, Btu/(lbm · °F)

Effect of Braking on Temperature



Rate of Heat Transfer

$$H_{\text{loss}} = \hbar_{\text{CR}} A(T - T_\infty) = (h_r + f_v h_c) A(T - T_\infty) \qquad (16\text{–}57)$$

where H_{loss} = rate of energy loss, Btu/s

$\hbar_{\text{CR}}$ = overall coefficient of heat transfer, Btu/(in^2 · s · °F)

h_r = radiation component of $\hbar_{\text{CR}}$, Btu/(in^2 · s · °F), Fig. 16–24*a*

h_c = convective component of $\hbar_{\text{CR}}$, Btu/(in^2 · s · °F), Fig. 16–24*a*

f_v = ventilation factor, Fig. 16–24*b*

T = disk temperature, °F

T_∞ = ambient temperature, °F